



Student Academic Management System

Cluster-based course enrolment with MySQL, Node.js and React

Team ID: 15

Section: 4

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Abstract

- A role-based **Academic ERP portal**: students pick a cluster, enrol and track grades; admins manage clusters and records.
- Built with **React**, **Node.js / Express** and **MySQL**: transactions, triggers, views and JWT security.

22

Tables

4

Clusters

38

REST APIs

2

Triggers



Problem Statement & Objectives

Problem: Manual registration causes over-booked seats, mixed sections and no audit trail.

- 1 Normalised MySQL schema with triggers, views & indexes
- 2 Cluster-locked, transaction-safe enrolment with waitlist
- 3 Secure REST API (JWT) with a React portal for both roles

Scope: Student & Admin · Clusters, Enrolment, Timetable, Grades



Existing vs Proposed System

Feature	Existing	Proposed
Registration	✗ Manual sheets	✓ Cluster-locked, online
Seat handling	✗ Over-booking	✓ Transactions + waitlist
Security	✗ Open records	✓ JWT + hashing
Audit trail	✗ None	✓ Trigger-based audit log
Gap	✗ No admin tools	✓ CGPA & cluster tools



Feasibility Study



Technical

JavaScript + SQL skills



Economic

100% open-source stack



Operational

Simple role-based UI



Schedule

Done within the semester



Tools & Technologies

Front-End

React 18

Vite

React Router

Back-End

Node.js

Express

REST APIs

Database

MySQL 8

Tools

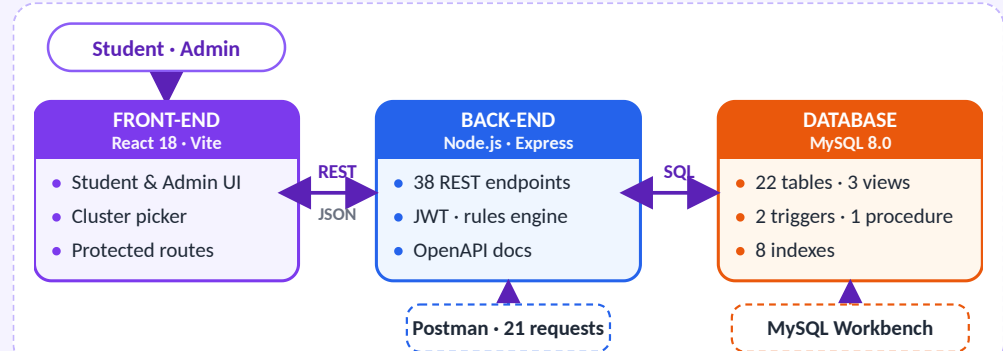
GitHub

Postman

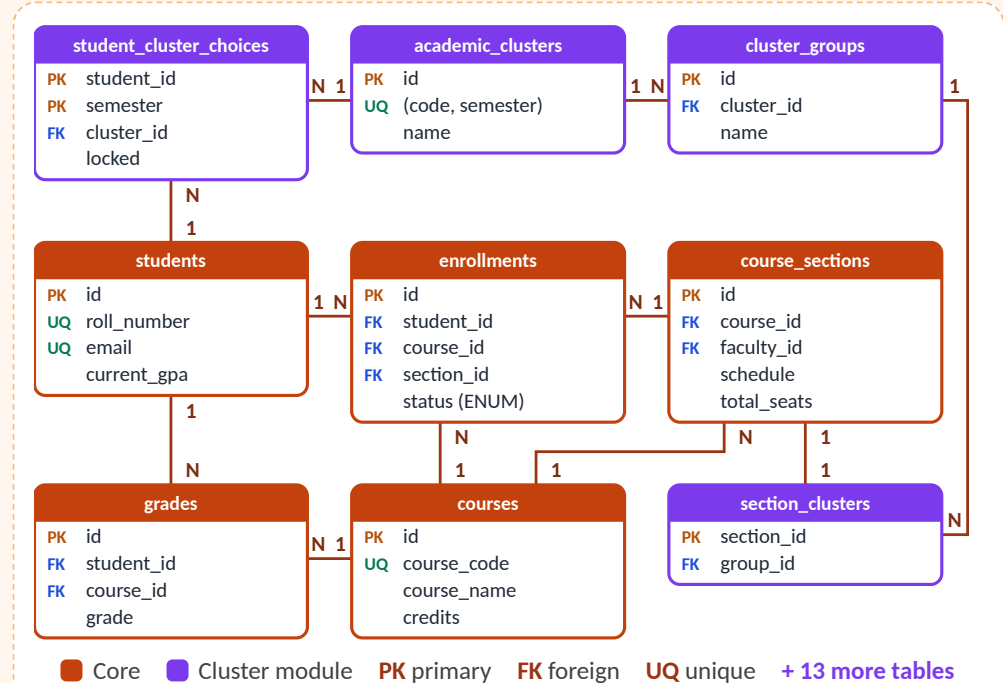
Workbench



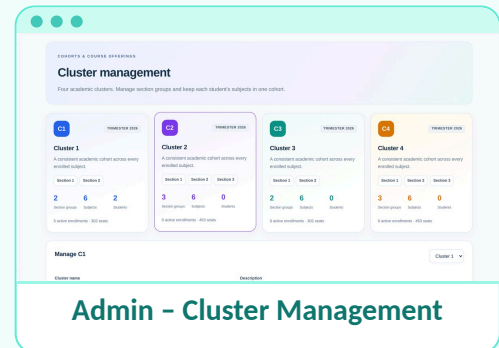
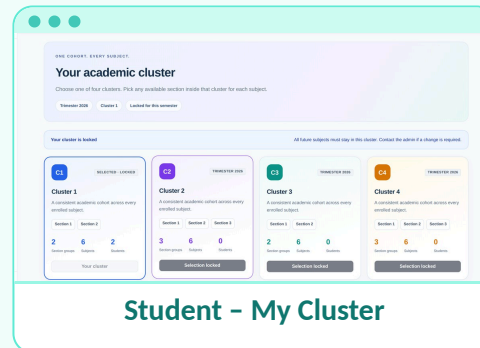
System Architecture



ER Diagram & Database Design



Implementation Screenshots



Results & Testing

- ✓ 7 unit + 40 integration checks pass
- ✓ Cross-cluster enrolment rejected; no seat lost
- ✓ Last-seat race admits exactly one student



Conclusion

- ✓ Working full-stack portal with cluster-locked enrolment on MySQL
- ✓ All three objectives achieved



Future Scope

Faculty login

Password recovery

Token revocation

Cloud deployment